

Functional diversity from network response dynamics

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Abstract We consider competing functional groups of tree species and develop a model of network response dynamics in order to measure the impacts of perturbations on the population distribution and diversity. The analysis of the equilibrium states relies on the connection between mean field game dynamics and replicator dynamics. We simulate our theoretical results from the data inventoried in French Guiana. Our results show that different types of disturbances modify the competitive interactions by affecting the evolutions of group densities. At the high regimes of disturbance, the canopy shade-intolerant species supplant the canopy shade-tolerant species. Tropical forest managers can thus take advantage of the competitive interactions between the functional groups to stimulate the abundance of marketable timber species. We also validate the hypothesis of maximum diversity at the intermediate disturbance levels.

Keywords Bioeconomics · Network response dynamics · Evolutionary games · Tropical forest management · Diversity

JEL Classification C61 · C65 · C7 · N5 · Q57

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1 Introduction

Forest dynamics depends on tree mortality, diameter growth, and recruitment. The multispecies forest dynamics, signifying the variability in composition, is usually based on four underlying assumptions.

First, the forest stand is considered to be stable and resilient, provided that increasing frequency of disturbances raises recovery rates, which indicates a degree of resilience in forests exposed to recurrent past disturbance (Cole et al. 2014). Second, the climate and the environment are presumed to be stable, mainly because the investigation of forest dynamics is useful under constant environment and management (Karev 2006). Third, the dynamics is assumed to be density-dependent, even if the role of density dependence in the population dynamics of tropical trees has been a subject of strong debate (Silva Matos et al. 1999). Indeed, it has been established that the density dependence occurs in tropical forests (Alvarez-Buylla et al. 1996) and has been argued that the former explains abundance and diversity in tropical forests (Janzen 1970; Connell 1971; Volkov et al. 2003). Forth, the species diversity results from disturbances, be it natural or silvicultural. As such, the species diversity is strongly linked to the forest stand dynamics (Favrichon 1998b). It can thus serve as an indicator of the intensity of disturbances (Huston 1979; Prévost and Sabatier 1993; Phillips et al. 1994). Forest disturbances are the natural or anthropic events that cause change in the structure and composition of a forest ecosystem. Natural disturbances include natural disasters and pathogens. Anthropic disturbances comprise logging, thinning or the introduction of invasive species.

Out of the 8 million ha of tropical forest that cover the French Guiana territory, the French National Forestry Office manages more than 5.3 million ha (INSEE 2004). The Office lists up to three hundred different tree species per hectare. Thereby, combining species into groups ultimately allows for better representation of the variability of individual behavior (Vancley 1991). Moreover, studying the functional groups of tree species has been proved to be of major significance in studying the forest dynamics (Chazdon et al. 2010; Barbosa et al. 2014). Considering the high interest attached to commercial species, which group should be separately handled when dealing with mixed-species forests (Buongiorno et al. 1994), Favrichon (1998a) reduces the species diversity down to five functional groups with similar behaviors. The groups have been qualified as understory shade-tolerant (Group 1), canopy shade-tolerant (Group 2), emergent mid-shade-tolerant (Group 3), canopy shade-intolerant (Group 4) and pioneer shade-intolerant (Group 5). Two functional groups are of commercial and economic importance in French Guiana. Most of the species placed on the market originate from Group 3, that is, the species which are able to reach the canopy and are characterized by a medium wood density, and from Group 4, which corresponds to long-lived softwoods.

Although the complex process of competitive intransitivity or nonhierarchical competition has been put forward (Laird and Schamp 2006, 2009; Ulrich et al. 2014), the functional groups are considered to be reciprocal, in the sense that the effect of A on B is identical to that of B on A (Fowler 1981). As a matter of fact, in a setting with more than three species, competition can range from perfect intransitivity to a perfect hierarchy, with a series of possible intermediate scenarios. Given this complexity, and

because intransitivity can reverse the hierarchy, we consider that the effects end up being proportional. Put differently, at the average rank order, the effects are identical by cause of the multitude of intermediate scenarios, which end up counterbalancing the locally identified competitive hierarchy.

Two dissimilar paradigms serve as an explanation to the patterns of community assembly in populations. Under the equilibrium theory, species dynamics are ruled by deterministic processes in relation to their surrounding environment (Tilman 1982). On the contrary, neutral theory predicts that the community composition is determined by stochastic events, historical incidents, and random dispersal events (Hubbell 2001). When studying disturbed environments, in which recovery from a magnitude of disturbance has critical implications (Chazdon 2008; Harvey et al. 2008), clarifying the pressure of these mechanisms is crucial. To differentiate between the two, we stand behind the results by Norden et al. (2009), who support the equilibrium paradigm.

While competition is a key factor driving stand dynamics (Franklin et al. 1987), climate is often considered to be preponderant. Notwithstanding, using a data set of long-term observations from undisturbed natural forests in Western Canada, Zhang et al. (2015) found that competition was the primary factor causing the long-term changes in forest stands, followed by the well-known and proven climate factors. We focus on the competition effects, for we know that facilitative effects rarely occur in the absence of predominant competition (Callaway 2007).

The mathematical formalization of forest stands serves to simulate their behavior over time and becomes a tool for both research and management (Vancley 1995). In line with Buongiorno and Michie (1980) and Buongiorno et al. (1995), the matrix model by Favrichon (1998a, b) is an elegant forest dynamics model, aggregated at the functional group level, having been the subject of anthropic disturbances. The author ends the discussion over his results by opening an avenue of research which falls within the theoretical behaviors of multispecies models. In an attempt to address his call, and in consonance with this approach, we decide to model the population dynamics of functional groups on the basis of inter-group competition and density. Our approach can be justified by the acknowledgment that the tropical forest ecosystems depend on the species interaction networks, and the deduction of its rules is known to be a complex problem (Azaele et al. 2006). The approach is all the more important, for sustainable tropical forest management is the subject of strong interest.

Ulrich et al. (2014) have recently proposed a statistical framework, built upon the transition matrices which describe the outcomes of species interactions, and applied it in a Markov chain model.¹ Much as their approach is relevant for explaining the colonization dynamics, especially when one single species can occupy one patch at a time, their model discards non-ergodic Markov chains. Likewise, the stationary assumption requires that the probability of any event only depends on the initial state instead of the history. Given our focus on the competition effects, which can be null, the inference drawn from the history of growth, and a presence of well-mixed population evolving in a spatially homogeneous framework, the implementation of replicator dynamics proves to be suitable.

¹ Leslie (1945) and Usher (1976) matrices also follow the rationale of the Markov chains.

Following the work by [Vandermeer and Yitbarek \(2012\)](#), we employ a competition effect issued from a network framework, which we build upon the average diameter growth of a functional group against those of other functional groups, the whole being weighted by the population distribution of groups.² This methodology enables us to incorporate mortality and recruitment, for they impact the population relative densities. In the interest of formalizing the dynamics as a variant of the replicator equation ([Hofbauer and Sigmund 1998](#)), we assume a spatially well-mixed species population.³ For the purposes of the article, we term it network response dynamics.

Indeed, network dynamics have become an increasingly used tool to represent interactions. For example, by combining a game theoretical framework with graph theory and dynamical systems ([Fisher and Reeves 1995](#); [Chawanya and Tokita 2002](#)), [Allesina and Levine \(2011\)](#) showed, through a network representation of competition, that the latter is a trigger to species diversity.

We consider large populations of interacting functional groups and develop a model of network response dynamics in order to measure the impacts of perturbations on the species diversity. We represent the mean behavior of species in well-mixed populations, such that functional groups are considered to be in a uniformly distributed spatial competition. Of course, these simplifications ought to impact the model outputs. Nevertheless, at the population level, they closely reflect the mean dynamics which we try to uncover. As to how studying the equilibrium state of population dynamics, we draw on the recent connection between mean field game dynamics and replicator dynamics in the game theory literature ([Tembine et al. 2009](#); [Tembine 2011](#)). Indeed, such as pointed out by [Azaele et al. \(2006\)](#), there have only been a few studies on the dynamics and stability of tropical rain forests. Furthermore, [Doncaster \(2009\)](#) showed that the Lotka–Volterra equation, which is equivalent to the continuous replicator equation ([Bomze 1983](#)), is able to estimate the average competition coefficients at some density-dependent equilibrium. At last, we test our theoretical predictions through competition matrices and relative densities of functional groups, which we compute on the basis of inventory data issued from the Paracou experimental forest station, in French Guiana.

Our results show that different types of disturbances modify the competitive interactions by affecting the evolutions of group densities. For instance, high regimes of disturbance lead to an increase of the canopy shade-intolerant species to the detriment of the canopy shade-tolerant species. We also validate the presence of equilibrium dynamics at the functional group levels. The simulations that we carry out demonstrate that forest managers can take advantage of the competitive interactions to stimulate the abundance of commercial or noncommercial timber species. It implies that the functional-group competition can serve as a decision aid tool to orient the silvicultural strategies. We also find that the hypothesis of maximum diversity at the intermediate

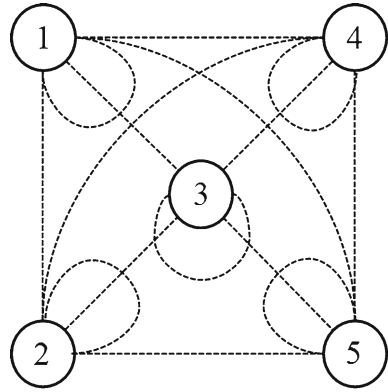
² Despite the fact that intra-species interactions can be substantial, [Doncaster \(2009\)](#) showed that the inter-specific impacts are to be considered as of equal magnitude to that of the intra-specific impacts. Besides, the self-loops in the graph implicitly integrate the intra-group competition effects.

³ Although the spatial geometry is important, segregation and aggregation are scale dependent, such as highlighted by [Hurlbert \(1990\)](#) and [Dale \(2000\)](#). As such, species might be segregated at a scale of a plot, but they are considered as aggregated at the experimental forest site.

Fig. 1 Schematic illustration of a competition network defined by a set of five nodes

$N = \{1, 2, 3, 4, 5\}$, and a set of 15 competitive interactions, five of which are self-loops

$C = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 2), (2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (3, 5), (4, 4), (4, 5), (5, 5)\}$



disturbance level is valid. Under certain conditions, and to a certain extent, sustainable tropical forest management—as an anthropic form of disturbance—can lead to more diversity.

After this starting section, we present the population dynamics model in Sect. 2. Section 3 is devoted to illustrating simulation examples. Section 4 discusses the results.

2 Competition network

Provided a set of connected functional groups forming a network, the topology of a networked forest system is encoded through a graph, such as shown in Fig. 1.

Let $G = (N, C)$ be an undirected graph, which reflects the reciprocity defined above, where $N = \{N_1, \dots, N_n\}$ is the set of nodes representing the densities of functional groups in which are distributed the numerous tree species. Let $C \subseteq N \times N$ be the set of edges or connections reflecting their competition.

The competition network model, such as the one pictured in Fig. 1, can be expressed as

$$\mathbf{C} = \mathbf{N} \cdot \mathbf{A} \quad (1)$$

where $\mathbf{N} = [N_1^T, \dots, N_n^T]^T$ corresponds to the state vector of the functional groups, with respect to the species relative abundance in ordination, and $\mathbf{A}^T = [a_{j,i}]$ to the matrix containing the competition payoff vectors, where $\sum_{j=1}^n a_{j,i}$ is the payoff of i against all j . The competition effect is normalized to $C_{i,j} \in [-1, 1]$. Thereby, the effect represents the network competitive response of a functional group to other functional groups. When $C_{i,j} \in [-1, 0)$, species belonging to i negatively respond to species belonging to j . Otherwise, the response is positive, for $C_{i,j} \in (0, 1]$,⁴ or independent, for $C_{i,j} = 0$. In the last case, the edge is absent.

This basic framework enables us to focus on how competitive interactions generate distinct population patterns. To explore those patterns, we now introduce the dynamics.

⁴ When $\lim(C_{i,j}) = 1$, we implicitly account for the facilitative or mutual effects.

2.1 Network response dynamics

By combining game theory and population dynamics in a variant of the replicator equation, we consider finite group populations, where $N_i \in [0, 1]$, which sum $\sum_{i=1}^n N_i$ must be equal to one, such that zero corresponds to a functional group extinction and one to a monotype population.

Let us now inject the competition effect into the population dynamics. Consider a population in which the tree species, called replicator, exist in several different types. Along the lines of the replicator dynamics, the time evolution of the functional group density $N(t)$ is given by the differential equation

$$\frac{dN_i}{dt} = N_i (1 - N_i) C_{i,j}. \quad (2)$$

We term it network response dynamics, for it mixes population dynamics in replication and network competition defined above. As a matter of fact, the dynamics is the weighted i -type payoff given the weighted average payoff in the population.⁵

2.2 Equilibrium state

The game theory literature has recently shown that replicator dynamics can be equated with mean field dynamics⁶ in population games with multiple types, for a class of evolutionary games can be obtained from the mean field games.

This analogy enables us to study the population equilibrium state, where the mean field limit can be seen as the population state. In this case, a population profile is an equilibrium if

$$N_i^* > 0 \Rightarrow \max_{N_i \in N} a_{j,i} (N_i^*). \quad (3)$$

This definition is equivalent to the variational inequality problem such that

$$\langle a_{j,i} (N_i^*), N_i^* - N \rangle \geq 0, \quad \forall N_i^* \in N. \quad (4)$$

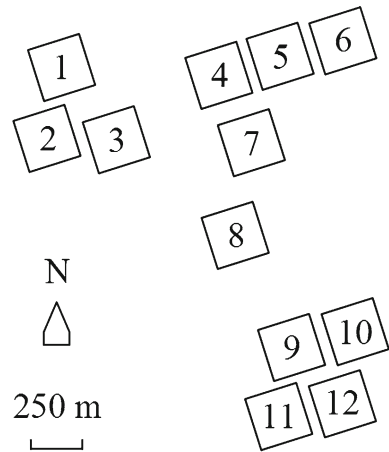
The interesting characteristic of the variational inequality problems is that they encompass the optimization problems, such that the network equilibrium state is also the solution to the network output optimization subjected to the competition effects.

If $a_{j,i}$ belongs to a continuous vector field, the population game has at least one equilibrium. In that case, the mean field game dynamics is positively correlated (Cheung 2014) if the natural bilinear functional verifies

⁵ In detail, we have $N_i \cdot \text{cov}(\sum_{i=1}^n N_i x_i, \sum_{j=1}^n N_j x_j) = N_i \cdot \sum_{i=1}^n N_i \cdot \sum_{j=1}^n N_j \cdot \text{cov}(x_i, x_j) = \sum_{i=1}^n N_i \cdot \sum_{j=1}^n (1 - N_i) \cdot N_i \cdot \text{cov}(x_i, x_j) = N_i \cdot (1 - N_i) \cdot \sum_{i=1}^n \sum_{j=1}^n N_i \cdot a_{j,i} = N_i (1 - N_i) C_{i,j}$.

⁶ According to this branch of game theory, the inter-type interactions must be sufficiently regular for the global phenomena to emerge.

Fig. 2 Positions of 12 square plots in the Paracou experimental forest site. Treatment 0: *Plot 1, Plot 6 and Plot 11*; Treatment 1: *Plot 2, Plot 7 and Plot 9*; Treatment 2: *Plot 3, Plot 5 and Plot 10*; Treatment 3: *Plot 4, Plot 8 and Plot 12*



$$\left\langle a_{j,i}(N_i), \frac{dN_i}{dt} \right\rangle > 0, \quad \forall N_i \in N. \quad (5)$$

Positive correlation means that, whenever the dynamics is not at rest, the inner product of the payoff vector and the motion vector is strictly positive.⁷ Under this property, the convergence occurs, that is, the population equilibrium state exists.

3 Simulations

The data we use to build the competition matrices and identify the relative densities of functional groups comes from the Paracou experimental forest station (5°15'N; 52°55'W), in French Guiana (Fig. 2). Over the time period going from 1984 to 2007, the database lists the annual diameter measure per tree, which amounts to more than 800,000 observations.

The primary forest is a tropical lowland moist evergreen forest with an extensive mixture of species. In 1984, 12 square plots of 6.25 ha were delimited. According to a randomized experimental design (Maître 1982), four silvicultural treatments or disturbances were applied.

Three undisturbed plots served as control plots of natural forest (Treatment 0). Three plots underwent a low-intensity logging in 1987 (Treatment 1). Three plots underwent a low-intensity logging in 1987, and a thinning by poison-girdling of non-commercial species in 1988 (Treatment 2). At last, three plots underwent an intensive

⁷ The rest points of the replicator equation are those for which all payoff values are equal (Sigmund 2011). The population game with payoffs $a_{j,i}$ is a full potential game if there is a continuous differential mapping $M : \mathbb{R} \rightarrow \mathbb{R}$, that we can use as the inner product of the payoff vector and the direction of motion, such that $dM(N_i)/dN_i = dN_i/dt$. The mean field game dynamics is positively correlated if $\dot{N}_i \neq 0 \Rightarrow \langle a_{j,i}(N_i), \dot{N}_i \rangle > 0$. On the contrary, the Euler stationarity would imply $\dot{N}_i = 0$ (Franc et al. 2000).

Table 1 Competition matrices

	$a_{1,i}$	$a_{2,i}$	$a_{3,i}$	$a_{4,i}$	$a_{5,i}$
(a) Treatment 0					
$a_{j,1}$	.00	.02	.02	(.00)	.00
$a_{j,2}$	.02	.09	.10	(.02)	.00
$a_{j,3}$	.02	.10	.15	(.02)	(.00)
$a_{j,4}$	(.00)	(.02)	(.02)	.05	.00
$a_{j,5}$	.00	.00	(.00)	.00	.01
(b) Treatment 1					
$a_{j,1}$	.00	.01	.01	(.03)	(.00)
$a_{j,2}$	.01	.55	(.20)	(1.11)	(.29)
$a_{j,3}$	.01	(.20)	.40	.20	.18
$a_{j,4}$	(.03)	(1.11)	.20	2.92	.59
$a_{j,5}$	(.00)	(.29)	.18	.59	.24
(c) Treatment 2					
$a_{j,1}$	.01	.01	.09	.01	.06
$a_{j,2}$	.01	.20	.23	.18	.26
$a_{j,3}$	.09	.23	2.18	1.43	1.09
$a_{j,4}$	.01	.18	1.43	2.42	.56
$a_{j,5}$	.06	.26	1.09	.56	.87
(d) Treatment 3					
$a_{j,1}$	.02	(.01)	.01	.21	.13
$a_{j,2}$	(.01)	.14	(.13)	(.57)	.02
$a_{j,3}$	.15	(.13)	1.36	2.16	.96
$a_{j,4}$	.21	(.57)	2.16	7.95	.86
$a_{j,5}$	.13	.02	.96	.86	1.06

logging in 1987, followed by fuelwood cuts, and finally a thinning by poison-girdling of noncommercial trees in 1988 (Treatment 3).

To test the effects of tree competition on the demographic rates, the commonly used indicator is the basal area (Zhang et al. 2015).⁸ Given the proportionality between basal area and tree diameter at breast height, and considering the data we have access to, we substituted between them for the sake of estimating, within the forest plots, the magnitude of competition between the functional groups.

3.1 Modus operandi

We first built the competition matrices (Table 1), for each treatment type, from the time frame of 23 years covered in the database.⁹ As explained in the theoretical section, to address the interannual variation, which affects the competitive relationships, the competition metric was chosen based on the analysis of covariance of the tree-growth (McDonald et al. 2002).

⁸ Basal area is the area occupied by the cross-section of tree trunks and stems at their base.

⁹ The figures in brackets are negative values.

Table 2 Competition effects

(a) Treatment 0	
$C_{1,j}$	.41
$C_{2,j}$	.19
$C_{3,j}$	(.03)
$C_{4,j}$	.00
$C_{5,j}$	.44
(b) Treatment 1	
$C_{1,j}$	.00
$C_{2,j}$	(.00)
$C_{3,j}$	.27
$C_{4,j}$	.55
$C_{5,j}$	.18
(c) Treatment 2	
$C_{1,j}$	.02
$C_{2,j}$	.08
$C_{3,j}$	.36
$C_{4,j}$	.34
$C_{5,j}$	.21
(d) Treatment 3	
$C_{1,j}$	.02
$C_{2,j}$	(.02)
$C_{3,j}$	.25
$C_{4,j}$	.57
$C_{5,j}$	.18

So as to obtain the competition effects of the functional groups, we then weighted the competition indices by the relative population densities (Table 2).¹⁰

The preceding operations enabled us to run the network response dynamics and to simulate the population restructuring in time. Figure 3 illustrates the simulations of group densities in undisturbed and disturbed stands starting from the latest inventory records.

Afterwards, the equilibrium states of the population dynamics were studied through the correlation between the competition effects and the distribution of functional groups.

Finally, we estimated the Piélou index (Piélou 1969). This index represents a standardization form of the Shannon entropy index, displaying the relations between the class frequencies and referring to the species evenness (Mulder et al. 2004).

3.2 Results

Our first ascertainment deals with the sameness between our simulations and the predictions made by Favrichon (1998b) in undisturbed stands. We observe a flat increase

¹⁰ For example, $C_{1,j} = N_1a_{1,1} + N_2a_{2,1} + N_3a_{3,1} + N_4a_{4,1} + N_5a_{5,1}$.

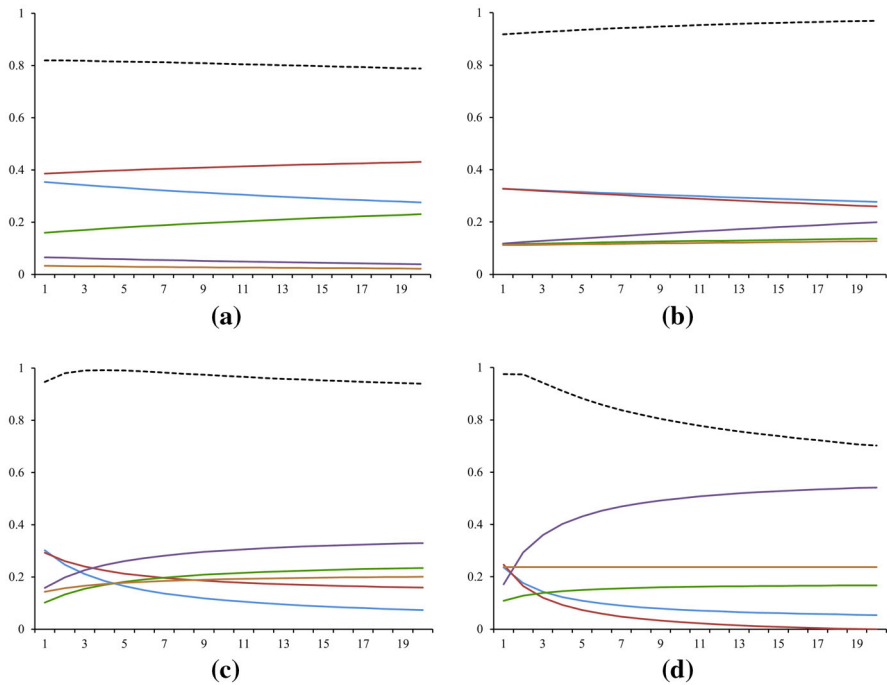


Fig. 3 Densities of functional groups which sum up to one (ordinates), as functions of time (abscissa) in undisturbed (Treatment 0) and disturbed (Treatment 1, Treatment 2 and Treatment 3) forest stands. Group 1 (blue curves): understory shade-tolerant species; Group 2 (red curves): canopy shade-tolerant species; Group 3 (green curves): emergent mid-shade-tolerant species; Group 4 (purple curves): canopy light-demanding species; Group 5 (brown curves): pioneer shade-intolerant species; Pielou index: diversity measure (black dashed curves). **a** Treatment 0. **b** Treatment 1. **c** Treatment 2. **d** Treatment 3. (Color figure online)

of Group 2 (11.47 %), which density goes from 0.39 to 0.43, as well as a flat decrease of Group 1 (−21.95 %), with a density passing from 0.35 to 0.27. Stable low densities of Group 4 and Group 5 can also be emphasized. The only mismatch relates to Group 3, which proportion goes up from 0.16 to 0.23 (43.94 %) in our case, whereas Favrichon (1998b) predicts an invariable level.

Some differences appear in Treatment 1. Although Group 1, Group 3 and Group 5 follow the same trajectories as in Favrichon (1998b), the simulations we carry out predict that the density of Group 4 should go upward (68.52 %) at the expense of that of Group 2 (−20.62 %). This rise can even be observed in the Favrichon (1998b) early predictions, but his simulations lead back to linear stability in time. In light of the fact that Group 4 belongs to the set of commercial species, our simulations reveal that the harvesting plan should favor their abundance.

With regard to Treatment 2 and Treatment 3, we obtain different results from Favrichon (1998b). In our case, they should accentuate the falls in densities of Group 1 (respectively −75.57 and −77.17 %) and Group 2 (respectively −45.46 and −100.00 %) and inflate the density growths of Group 3 (respectively 130.66 and 55.09 %) and Group 4 (respectively 108.03 and 216.20 %). As regards Group 5,

Treatment 2 enhances its relative density (39.59 %), while Treatment 3 maintains it unchanged (−0.26 %). Therefrom, two observations can be made. First, the silvicultural treatments of logging and thinning support the expansion of heliophilous species at the expense of shade-tolerant species. Second, as previously mentioned, these forestry works contribute to the overall spread of commercial species.

The connection between mean field game dynamics and replicator dynamics produces the following results. Through the measure of the linear relationship strength between payoffs and densities, the equilibrium states are validated for Treatment 0 (0.61, $p < 0.01$), Treatment 1 (0.66, $p < 0.01$), Treatment 2 (0.58, $p < 0.01$) and Treatment 3 (0.89, $p < 0.01$), for the correlation coefficients are positive. In all cases, the equilibrium state of the population dynamics is reached.

Subsequent to the simulations of population dynamics, we also calculated the magnitude of diversity through the Piélou index study. In spite of the theoretical reconfigurations within the forest stands, we can notice that Treatment 0, Treatment 1 and Treatment 2 maintain a high level of diversity. All the indices exceed the value of 0.8. Furthermore, Treatment 1 results in a rise in diversity (5.65 %), supporting the hypothesis of maximum diversity at the intermediate regimes of disturbance (Molino and Sabatier 2001), such as the low-intensity logging. Treatment 2 also yields an early increase, but displays a long-term pseudo bell-shaped form, thereby validating the hypothesis in an ephemeral manner. The only treatment which should provoke a decrease in evenness is Treatment 3 (−28.03 %). This result is consistent with the results obtained from a study conducted on forest logging and thinning by El-Kassaby and Benowicz (2000).

4 Discussion

Along with what has been previously shown by McDonald et al. (2002), we find that different treatments modify the competitive interactions by affecting the magnitude of tree-growth, which in turn updates the variations of group densities. In addition to providing a population distribution, we also give an analytical understanding of the species turnover in presence of disturbances.

With regard to the equilibrium states, while some of the literature contends that tropical forests are mega-diverse ecosystems that display complex and non-equilibrium dynamics (Fort and Inchausti 2013), our results show that, unlike the individual-based models, the networks built on functional groups lead to equilibrium dynamics. The results are thus consistent with the outcomes obtained by Azaele et al. (2006), who demonstrate that tropical ecosystems at stationarity are marginally stable, in that the convergence takes place while the dynamics is not at rest. As a consequence, the disturbed stands are not frozen in time, which suggests the possibility that they revert back to Treatment 0 over long periods.

Favrichon (1998b) formulates several concluding remarks by stating that intensive cutting (Treatment 3) should lead to the dominance of light-demanding softwoods. Despite their commercial interest, these species are considered to be risky, because they end up being unharvested when their wood density is too low. Likewise, the

author states that light cutting (Treatment 1 and Treatment 2) should be used to avoid this state of fact, ergo to allow for the production of high quality hardwood.

We partially confirm his assertions. We do support his predictions over the progressive spread of light-demanding softwoods, even in the case of low-intensity logging when the perturbation is accentuated with selective thinning.¹¹ And yet, although light-demanding softwoods may have a moderate commercial value, their traded volumes are significant, as has been recalled in the introduction. Besides, through the competition between the functional groups, our simulations show that the fraction of medium wood density species from Group 3 should also augment. These species are accounted for as part of the hardwood quality species. Provided that they just as much correspond to commercial species, their increase should provide the foresters with sufficient volumes demanded by the other market portion. Hence, the silvicultural strategies tested by forest managers could meet different portions of the market demand by yielding adequate volumes of production.

Furthermore, we know that the Amazonian rainforest is in general made up of many small-sized tree species (Group 1 and Group 5) which store little carbon, for they grow slowly and die before reaching full light. While canopy species are rarer (Group 3 and Group 4), they are the ones that store the most carbon (Fauset et al. 2015). Accordingly, not only would the tropical forest management increase the expansion of commercial species, but also could deliver beneficial effects in combating climate change.

Last but not least, the validation of the hypothesis, whereby the maximum diversity can be obtained at the intermediate regimes of disturbance, demonstrates that the tropical forest management—as an anthropic form of disturbance—and diversity can, to a certain extent, be strategically complementary.¹²

All in all, our results point out that forest management can theoretically redirect the tropical forest stands toward production, as long as they are not subject to deforestation, without necessarily compromising their diversity. The interesting implication of the framework we invested is that forest owners can use the competition effects of the functional groups to stimulate the abundance of commercial or noncommercial timber species. Hence, the analysis of the inter-group competition can be relevant, as a decision aid tool, to orient the silvicultural strategies and predict their economic outcomes, at least on a large scale.¹³

The obvious limit of the approach we propose falls within the lack of complexity of the competition matrices, which could be modulated by environmental triggers. Indeed, environmental conditions could allow for the co-occurrence of competing species (Connor and Simberloff 1983), while the sole prism of competition favors or disadvantages the fitness of a group of species. The extension of this work could also pertain to the introduction of a random graph, with stochastic competition matrices, so as to seize the impacts of climate change on the French Guiana tropical forest.

¹¹ These trees may simply respond to logging because they are more demanding of light (Poorter et al. 2005).

¹² In game theory, the decisions of agents are said to be strategic complements if they mutually increase their respective payoffs.

¹³ The same conclusion was empirically brought forward by Canham et al. (2006) within local neighborhoods, at the species level.

In the absence of biophysical multipliers, reflecting the probabilities of species presence conditional on climate variation, we encourage tropical forest ecologists to undertake projects in this direction and await their results with the purpose of updating the network response dynamics.

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